# CMSC 137

## *Data Communications and Networking*

Lecture Module

25 – Other Wireless Networks

University of the Philippines
LOS BAÑOS

# Module 25: Other Wireless Networks

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# Cellular Networks: Overview

# Overview

- Designed to provide communications between two moving units, called **mobile stations** or between one mobile unit and one stationary unit, called **land unit**

- A **service provider** must be able to locate and track caller, assign a channel to the call, and transfer the channel from base station to base station as the caller moves out of range

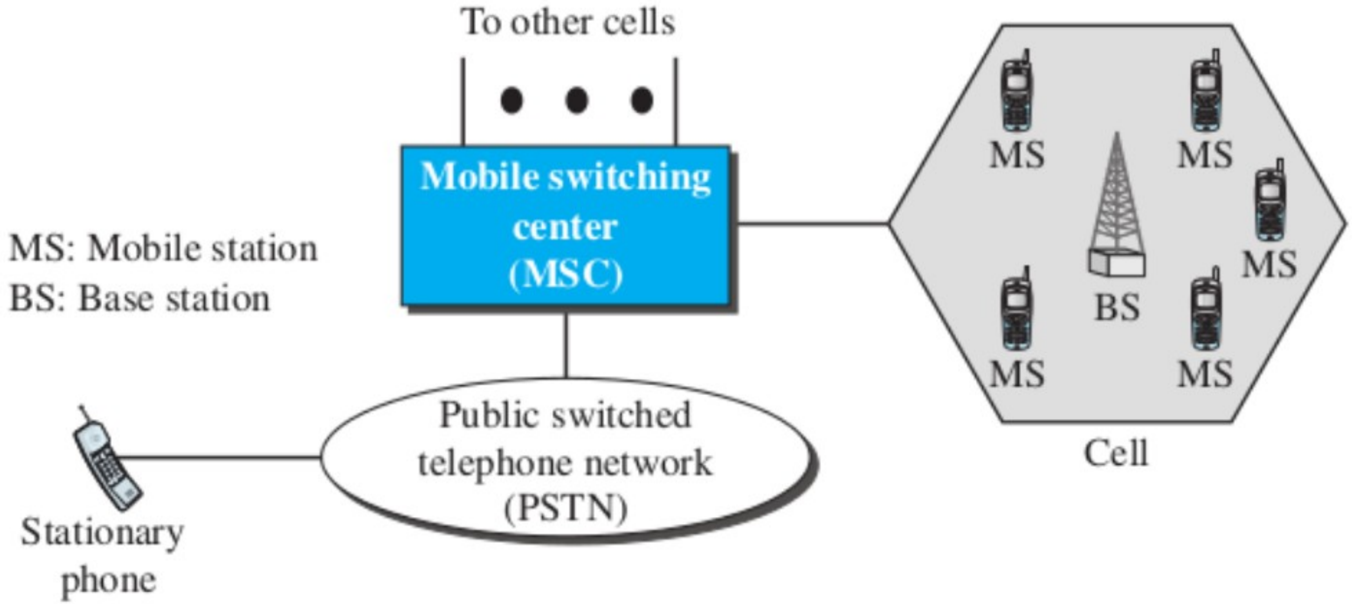
# Overview

- To enable tracking, cellular service area is divided into small regions called **cells**
- Each cell has an antenna powered by a network station, called the **base station**
- Each base station is controlled by a switching office, called a **Mobile Switching Center (MSC)** to coordinate communication between all base stations and the telephone central office

# Overview

- Cell size is not fixed, typical radius of 1 to 12 miles
- Cell size is optimized to prevent interference of adjacent cell signals
- Transmission power of each cell is kept low to prevent interfering with the signal

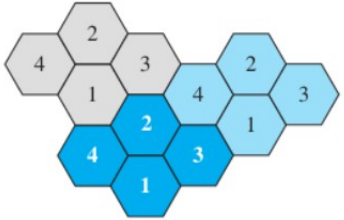
# Overview


To other cells
Mobile switching center (MSC)
MS: Mobile station
BS: Base station
MS
MS
MS
BS
MS
MS
Cell
Public switched telephone network (PSTN)
Stationary phone

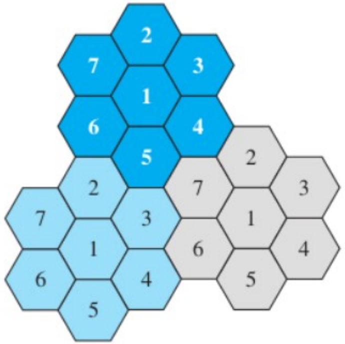
# Cellular Networks: Operation

# Frequency Reuse Principle

- Number of cells define the pattern, cells with same number in a pattern can use the same set of frequencies




a. Reuse factor of 4




b. Reuse factor of 7

# Transmitting

- Caller enters a code of 7 or 10 digits (**mobile number**)
- Mobile stations scans the band seeking a **setup channel** with a strong signal and sends the phone number to the closest base station using the channel
- Base station relays the mobile number to the MSC, MSC sends data to the telephone central office
- If called party is available, MSC assigns an unused voice channel to the call and the connection is established

# Receiving

- Telephone central office sends the number to the MSC
- MSC searches for the location of the mobile station by sending query signals to each cell, called **paging**
- Once the mobile station is found, MSC transmits a ringing signal
- When mobile station answers, voice channel is assigned to the call allowing voice communication to begin

# Handoff

- During conversation, mobile station may move from one cell to another – signal may become weak
- MSC monitors signal level every few seconds, if signal is weak MSC finds a new cell then transfers the call to the new channel
- **Hard handoff** – mobile station only communicates with one base station
- **Soft handoff** – mobile station can communicate to both base stations in the old and new cells

# Roaming

- Service provider has limited coverage
- Neighboring service providers can provide extended coverage through a **roaming contract**

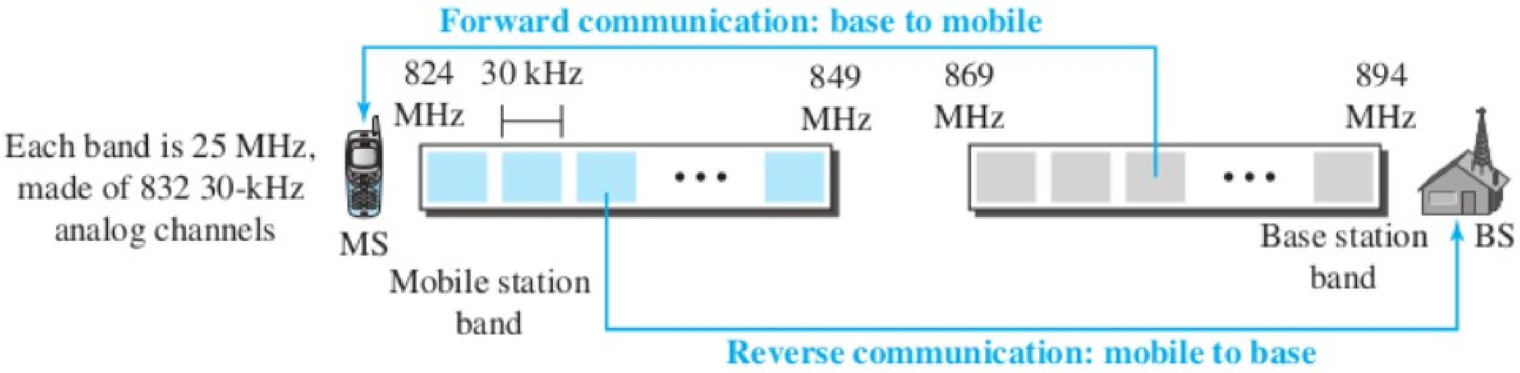
# Cellular Networks:
# First Generation (1G)

# AMPS

- **Advanced Mobile Phone System (AMPS)** – used in North America, uses FDMA
- Designed for voice communication using analog signals
- Uses ISM 800-MHz band
  - **Forward channel** (base to mobile) - (869-894 MHz)
  - **Reverse channel** (mobile to base) - (824-849 MHz)

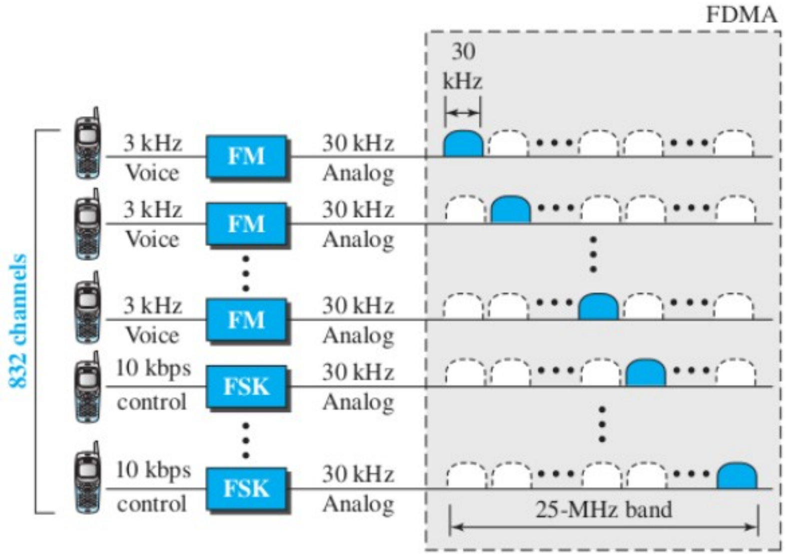
# AMPS

- 21 channels are used for control
- Frequency reuse factor is 7



# AMPS

- Uses FM(voice channel) and FSK(control channel)



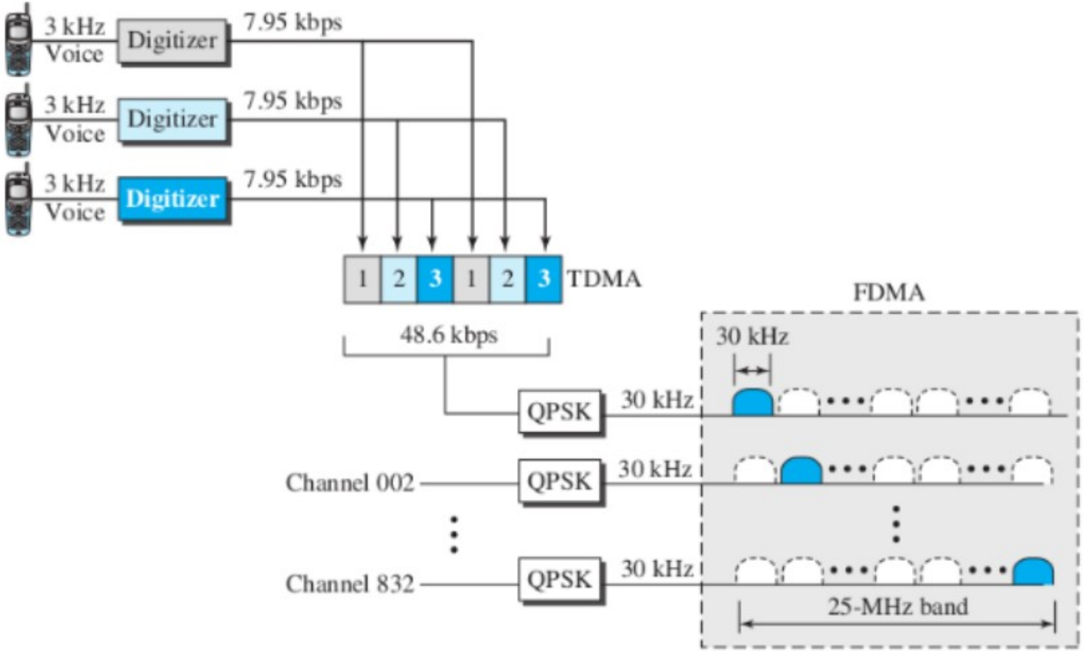
# Cellular Networks:
# Second Generation (2G)

# 2G

- Designed for digitized voice communications
- Systems: D-AMPS, GSM, IS-95

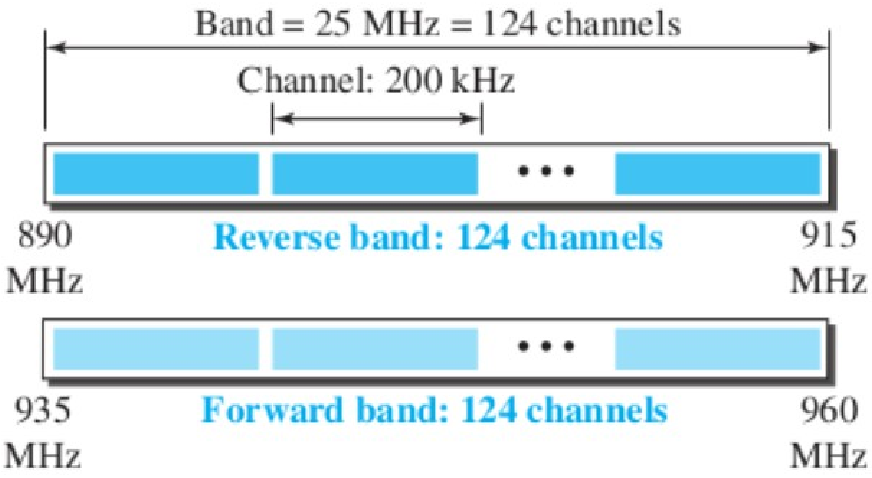
# D-AMPS

- 

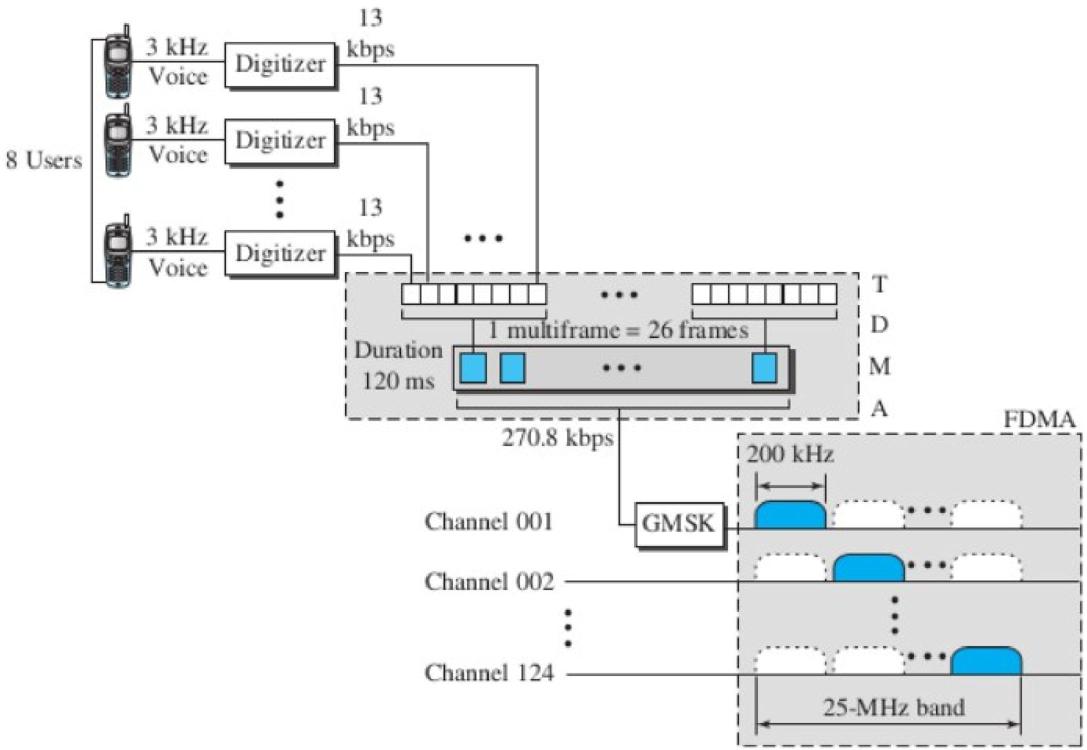

3 kHz
Voice
Digitizer
7.95 kbps
3 kHz
Voice
Digitizer
7.95 kbps
3 kHz
Voice
Digitizer
7.95 kbps
1
2
3
1
2
3
TDMA
48.6 kbps
QPSK
30 kHz
FDMA
30 kHz
Channel 002
QPSK
30 kHz
Channel 832
QPSK
30 kHz
25-MHz band

# GSM

- **G**lobal **S**ystem for **M**obile Communication (GSM) – Europe-based
- Reuse factor: 3



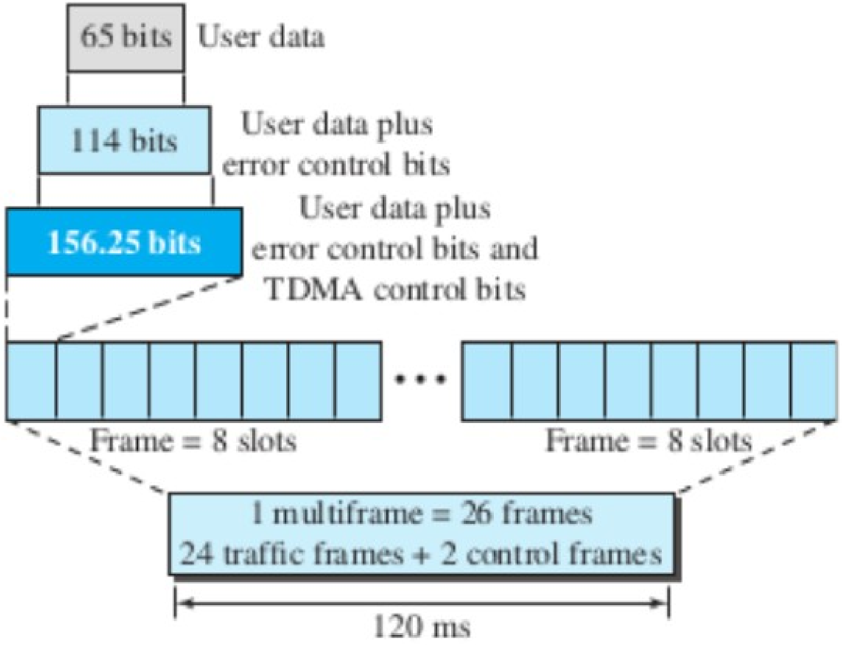
# GSM



8 Users
3 kHz
Voice
Digitizer
13
kbps
3 kHz
Voice
Digitizer
13
kbps
3 kHz
Voice
Digitizer
13
kbps
T
D
M
A
1 multiframe = 26 frames
Duration
120 ms
270.8 kbps
FDMA
200 kHz
Channel 001
GMSK
Channel 002
Channel 124
25-MHz band

# GSM

- Overhead



# Cellular Networks:
# Third Generation (3G)

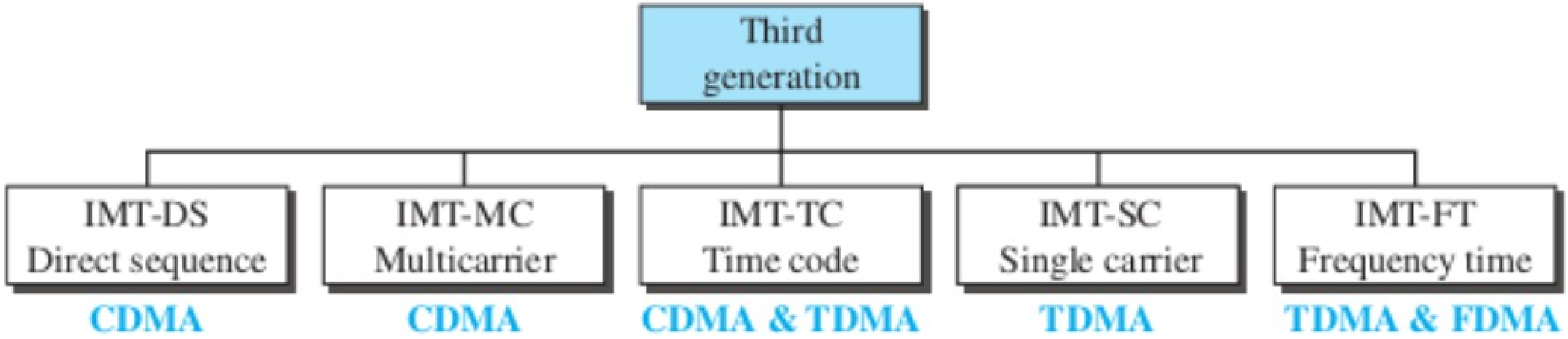
# 3G

- Provide both digital data and voice communication
- Started in 1992, Internet Mobile Communication 2000 (IMT-2000)

# IMT-2000 Blueprint

- Voice quality comparable to that of the existing PSTN
- Data rate of 144 kbps for moving car, 384 kbps for pedestrians, 2 mbps for office
- Support for **packet-switched** and **circuit-switched** data services
- A band of 2 GHz
- Bandwidths of 2 MHz
- Interface to the Internet

# IMT-2000 Radio Interfaces



# Cellular Networks:
# Fourth Generation (4G)

# 4G Objectives

- Spectrally efficient system
- High network capacity
- 100Mbps for moving car, 1Gbps for stationary users
- Smooth handoff across heterogeneous networks
- Seamless connectivity and global roaming
- All IP, packet-switched, networks

# 4G Features

- Access Scheme
  - Orthogonal FDMA (UMTS), Interleaved FDMA (UMTS), Multicarrier CDMA (IEEE 802.20)
- Modulation
  - 64-QAM for LTE
- Radio System
  - Software Defined Radio (SDR)
- Antenna
  - Multiple-input Multiple-output (MIMO), Mutiuser MIMO

# Cellular Networks: Fifth Generation (5G)

# 5G Features

- Uses Orthogonal FDM encoding
- Uses millimeter waves (wavelength)
- limited range, can be blocked by materials like walls and windows
- health concerns?
- Can use low, medium, or high frequency bands (high frequency bands, 24-47 GHz, can achieve peak at 20 Gbps data rate)
- Internet of Things
- Useful for urban areas

# 5G

https://www.nperf.com/en/map/PH/-/1999180.SmartSunTNT-Mobile/signal?ll=14.180933667524748&lg=121.24717712402345&zoom=14

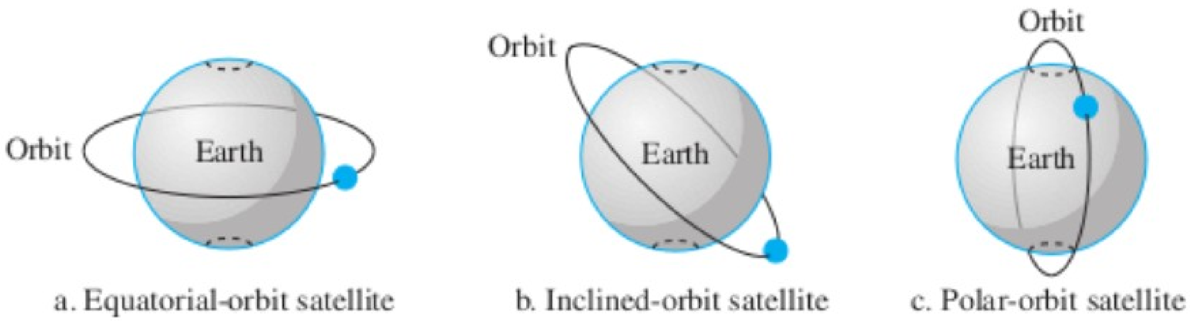
# Satellite Networks

# Overview

- Combination of nodes: **artificial satellites**, **Earth stations**, **end-user terminals or phones**
- Similar to cellular networks, Earth is divided into cells
- Useful in underdeveloped regions

# Orbits

- Period – time required for a satellite to make a complete trip around Earth
  - Period = 1/100 * distance 1.5
  - Earth's radius is 6378 Km



# Footprint

- Process microwaves with bidirectional antennas (line-of-sight)
- **Footprint** is the specific area where the signal is aimed and the signal strength is maximum

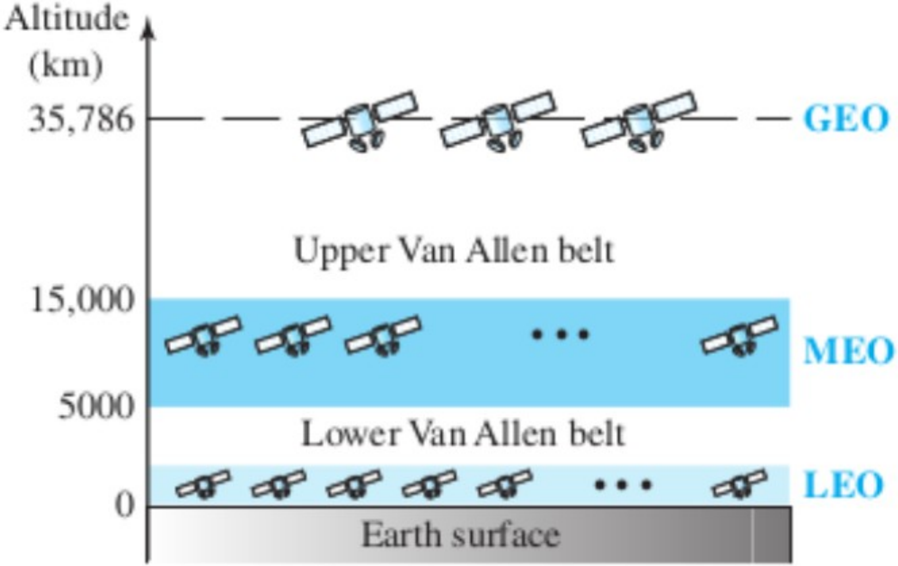
# Frequency Bands

| *Band* | *Downlink, GHz* | *Uplink, GHz* | *Bandwidth, MHz* |
|---|---|---|---|
| L | 1.5 | 1.6 | 15 |
| S | 1.9 | 2.2 | 70 |
| C | 4.0 | 6.0 | 500 |
| Ku | 11.0 | 14.0 | 500 |
| Ka | 20.0 | 30.0 | 3500 |

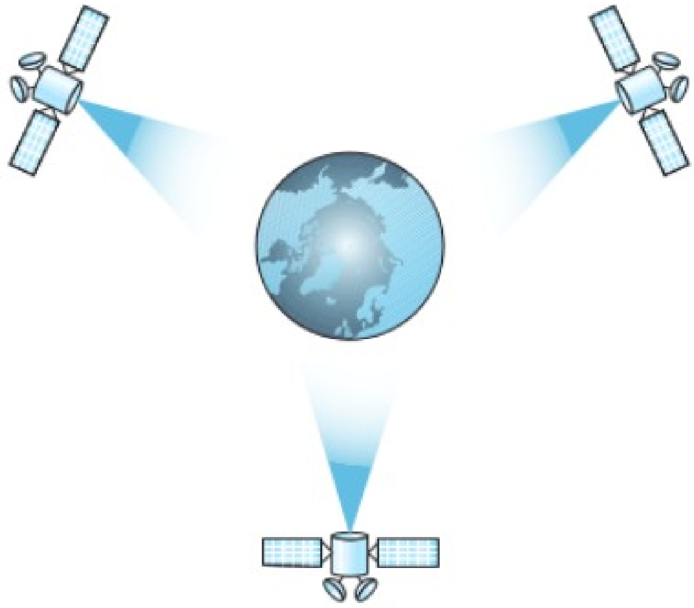
# Satellite Categories

- **Geostationary Earth Orbit (GEO)** – 35786 Km
  - Typical satellite networks
- **Medium Earth Orbit (MEO)** – 5000-15000 Km
  - Example GPS
- **Low Earth Orbit (LEO)** – below 2000 Km

# Satellite Categories

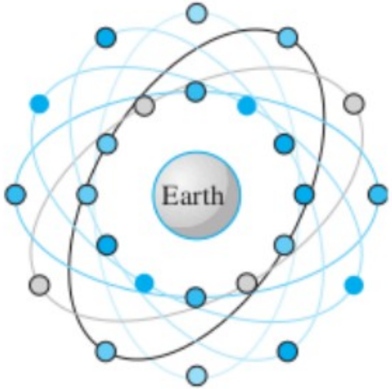


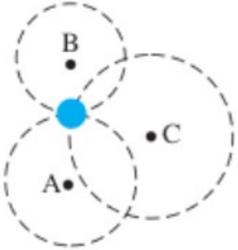
# GEO



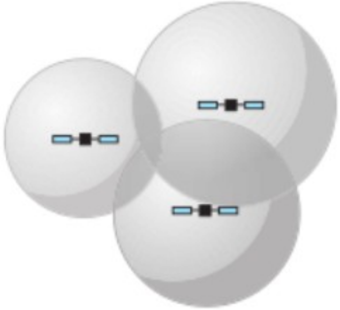
# MEO

- Global Positioning System




a. Two-dimensional trilateration

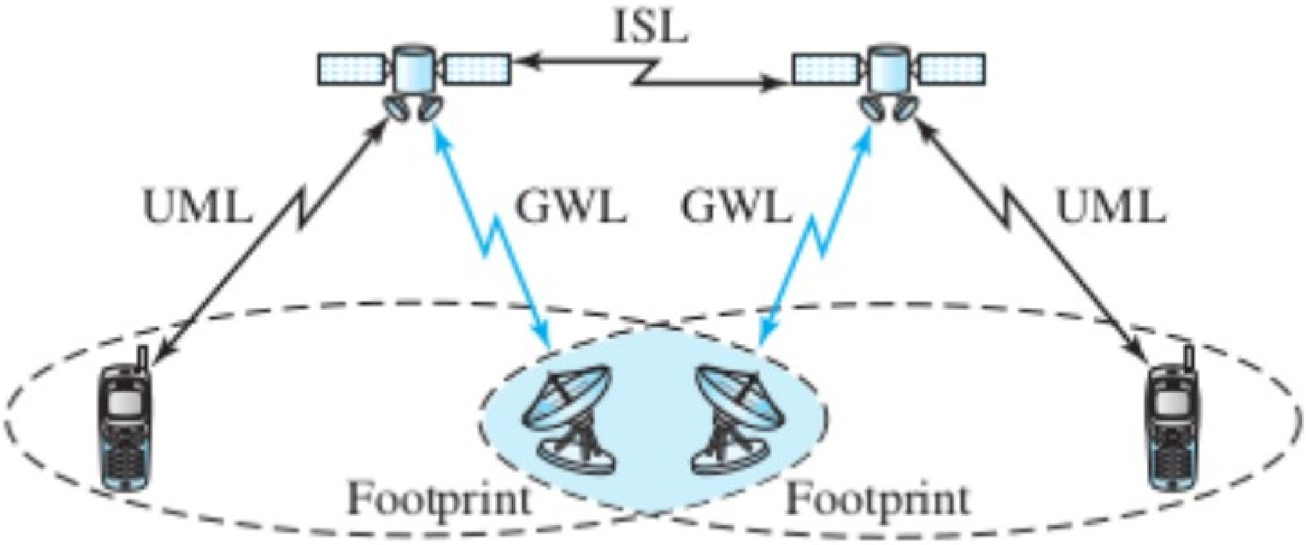


b. Three-dimensional trilateration

# LEO

- Altitude between 500 and 2000 Km
- Satellite Speed: 20000 to 25000 Km/h
- Footprint diameter: 8000 Km
- Uses a constellation of Satellites connected with each other through **intersatellite links (ISL)**
- Mobile system connects through a **user mobile link (UML)**
- Satellite communicate with Earth station using a **gateway link (GWL)**

# LEO



# Starlink

https://www.starlink.com/technology

# Starlink

https://satellitemap.space/?constellation=starlink